ONLINE MASTERS IN **DATA SCIENCE**

DSC 258 – NATURAL LANGUAGE PROCESSING

NEURAL N-GRAM MODELS

UC San Diego

COMPUTER SCIENCE & ENGINEERING

HALICIOĞLU DATA SCIENCE INSTITUTE

Neural N-gram Model

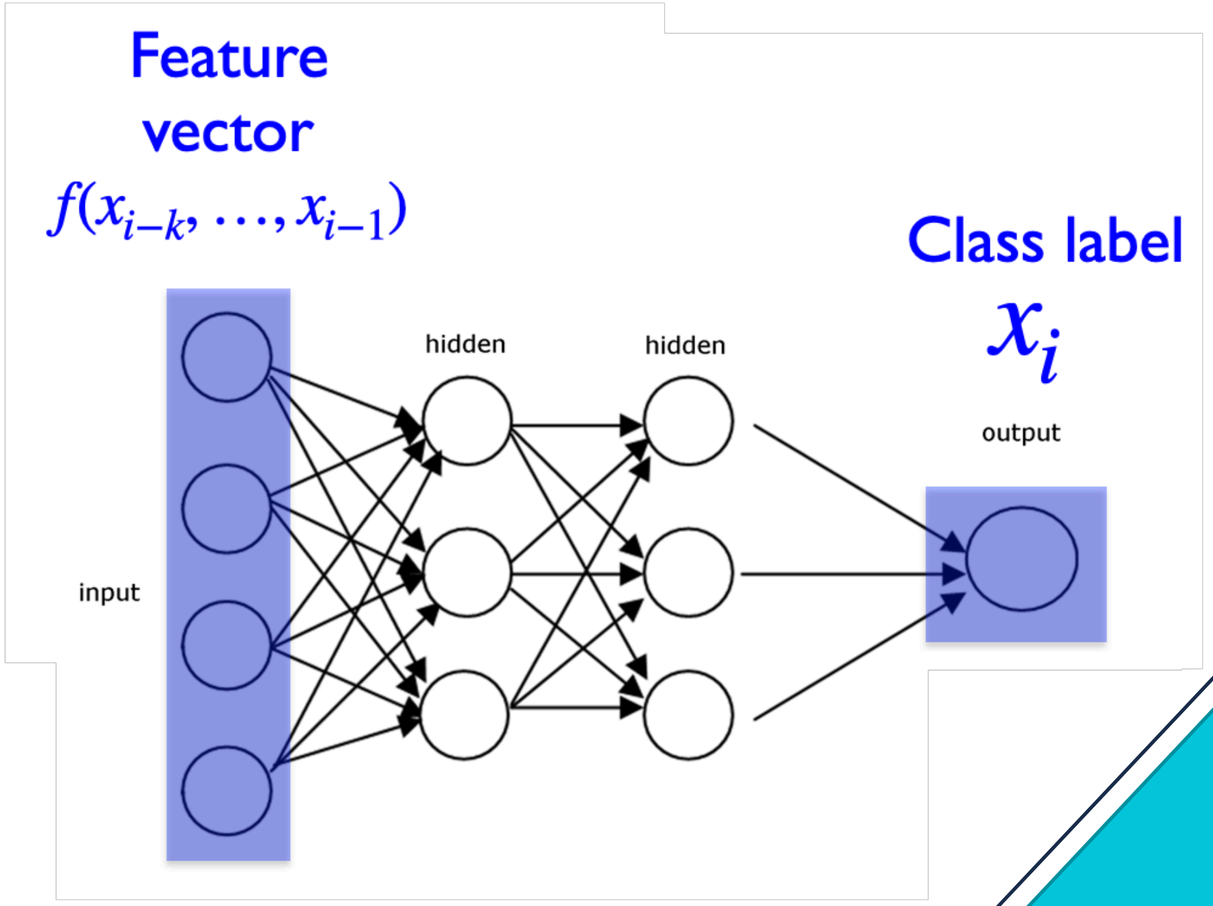
$$p(x_i \mid x_{i-k}, \ldots, x_{i-1})$$

Logistic regression:

$$\propto \exp(w^\top f(x_{i-k}, \ldots, x_{i-1}))$$

Fully-connected NN:

$$\propto \exp w_3^\top \sigma W_2^\top \sigma W_1^\top f(x_{i-k}, \ldots, x_{i-1})$$



Neural Ngram Model

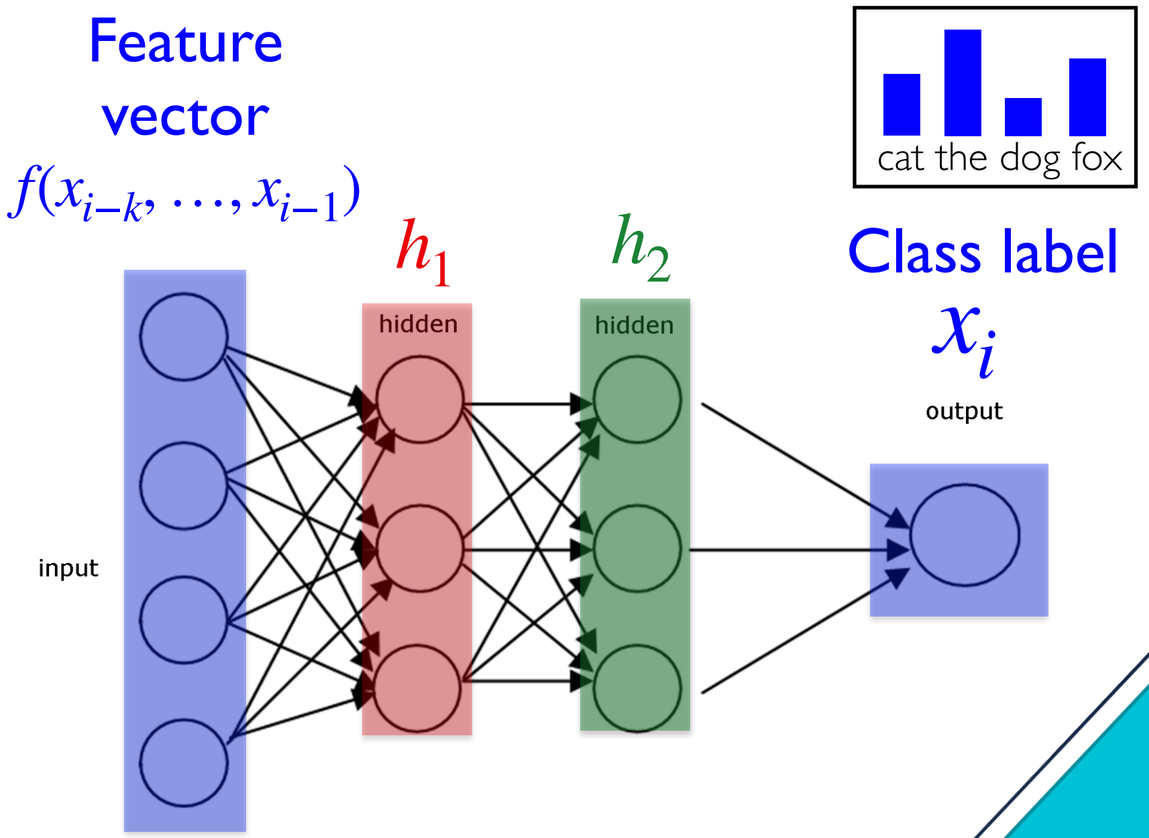
$$p(x_i \mid x_{i-k}, \ldots, x_{i-1})$$

$$\propto \exp w_3^\top \sigma W_2^\top \sigma W_1^\top f(x_{i-k}, \ldots, x_{i-1})$$

Hidden layer 1: $h_1 = \sigma W_1^\top f(x_{i-k}, \ldots, x_{i-1})$

Hidden layer 2: $h_2 = \sigma W_2^\top h_1$

Output layer: $p(x_i \mid \ldots) \propto \exp w_3^\top h_2$



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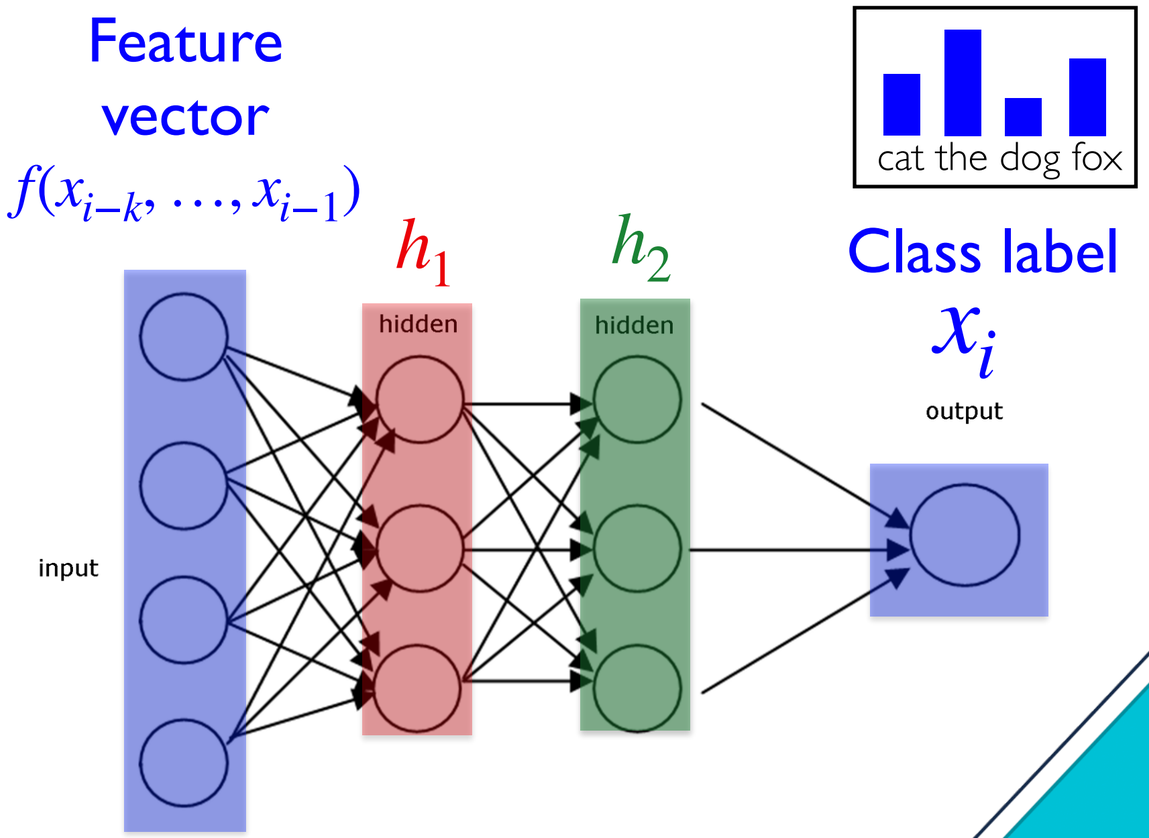

Neural Ngram Model

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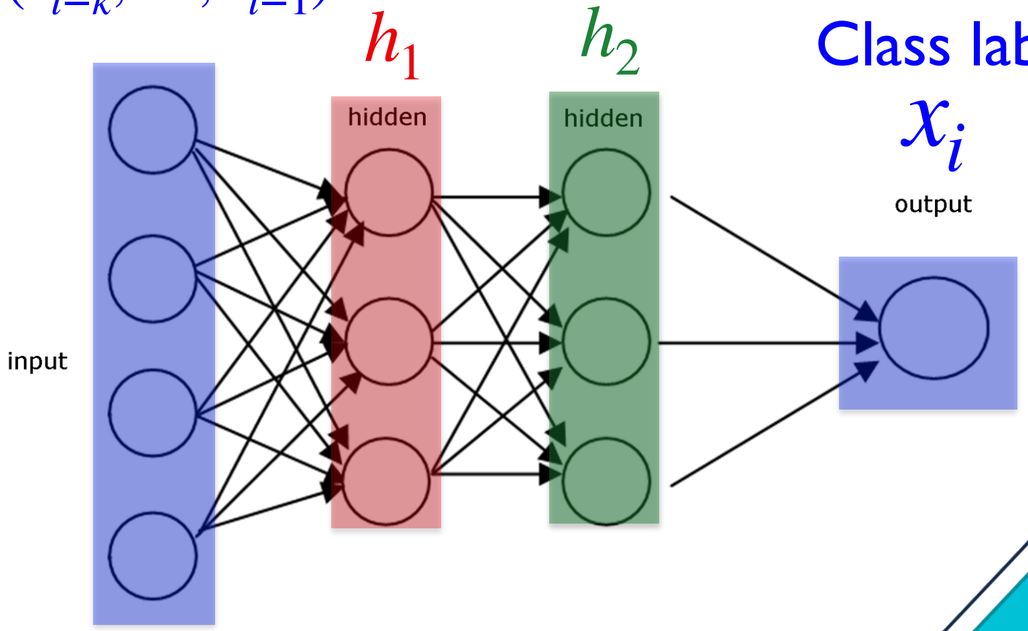
$$\propto \exp w_3^\top \sigma W_2^\top \sigma W_1^\top f(x_{i-k}, \ldots, x_{i-1})$$

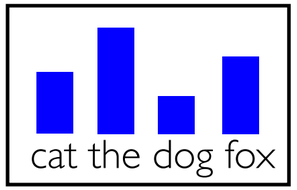
$$\theta$$

Model parameters

Feature vector

$f(x_{i-k}, \ldots, x_{i-1})$





Class label

x_i

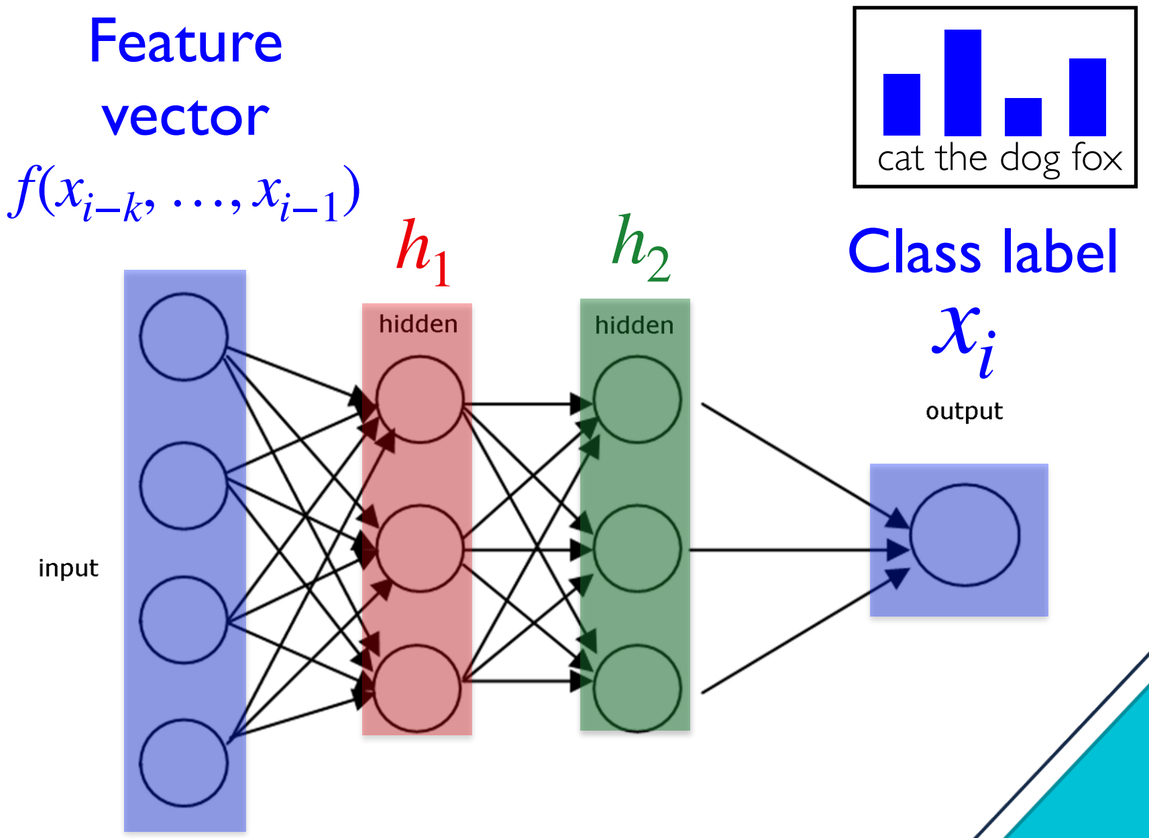
Neural Ngram Model

$$p(x_i \mid x_{i-k}, \ldots, x_{i-1})$$

$$\propto \exp w_3^\top \sigma W_2^\top \sigma W_1^\top f(x_{i-k}, \ldots, x_{i-1})$$

Training objective

$$\max_\theta \sum_i \log p_\theta(x_i \mid x_{i-k}, \ldots, x_{i-1})$$



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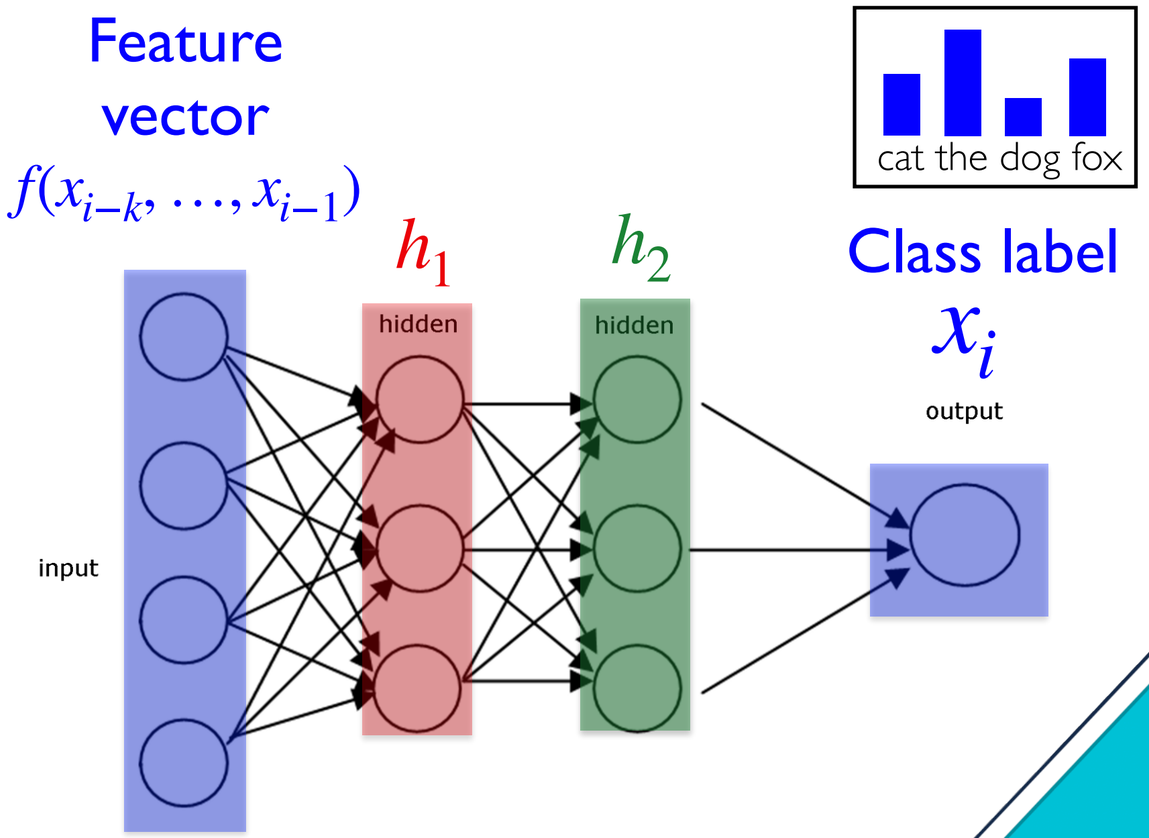
$$\propto \exp w_3^\top \sigma W_2^\top \sigma W_1^\top f(x_{i-k}, \ldots, x_{i-1})$$

Gradient ascent!

$$\nabla_\theta \sum_i \log p_\theta(x_i \mid x_{i-k}, \ldots, x_{i-1})$$



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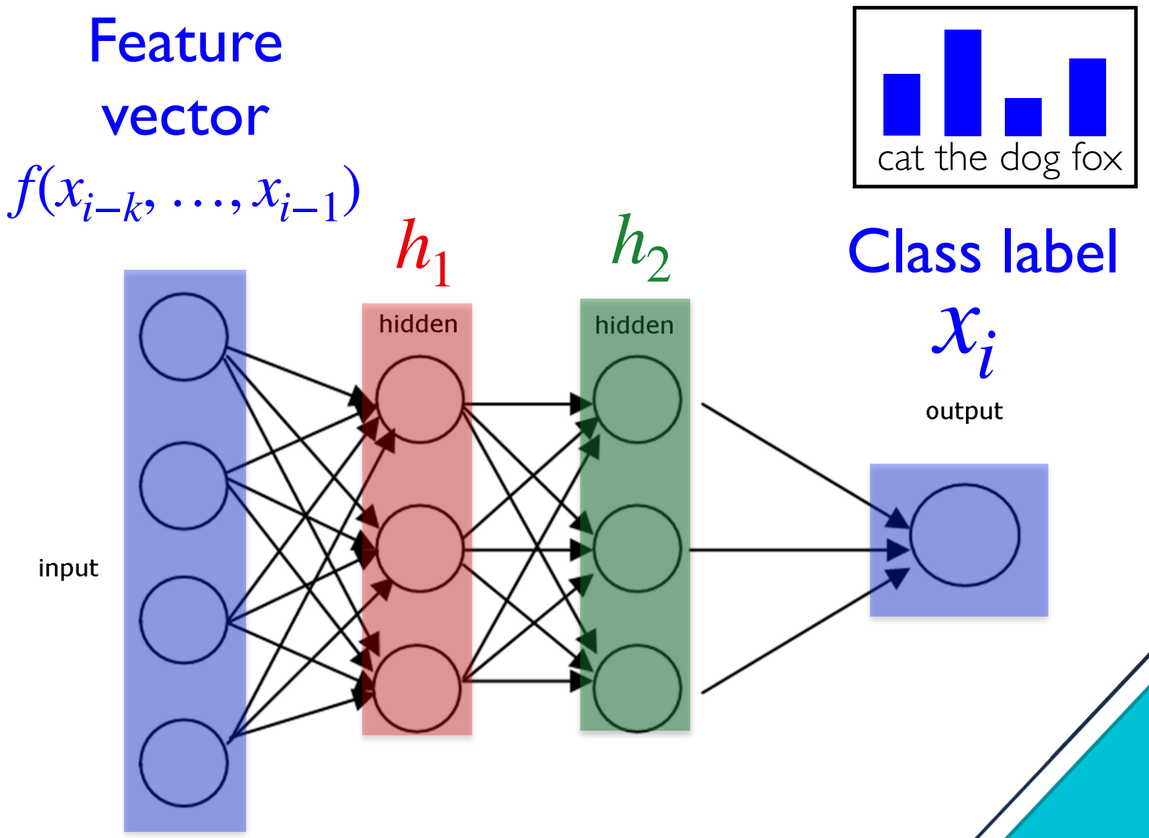
Stochastic gradient ascent!

$$\nabla_\theta \ \log p_\theta(x_i \mid x_{i-k}, \ldots, x_{i-1})$$

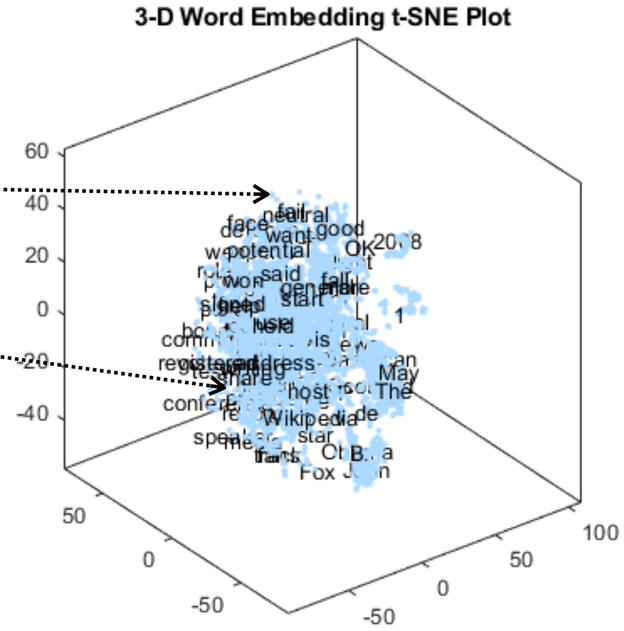


Input features?

$$f(x_{i-k}, \ldots, x_{i-1}) = concate(x_{i-k}), \ldots, e(x_{i-1})$$

e(the)

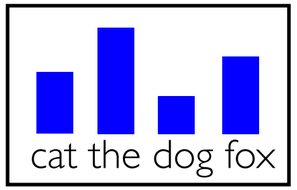
e(cat)



$$p(x_i \mid x_{i-k}, \ldots, x_{i-1})$$

Feature vector

$f(x_{i-k}, \ldots, x_{i-1})$



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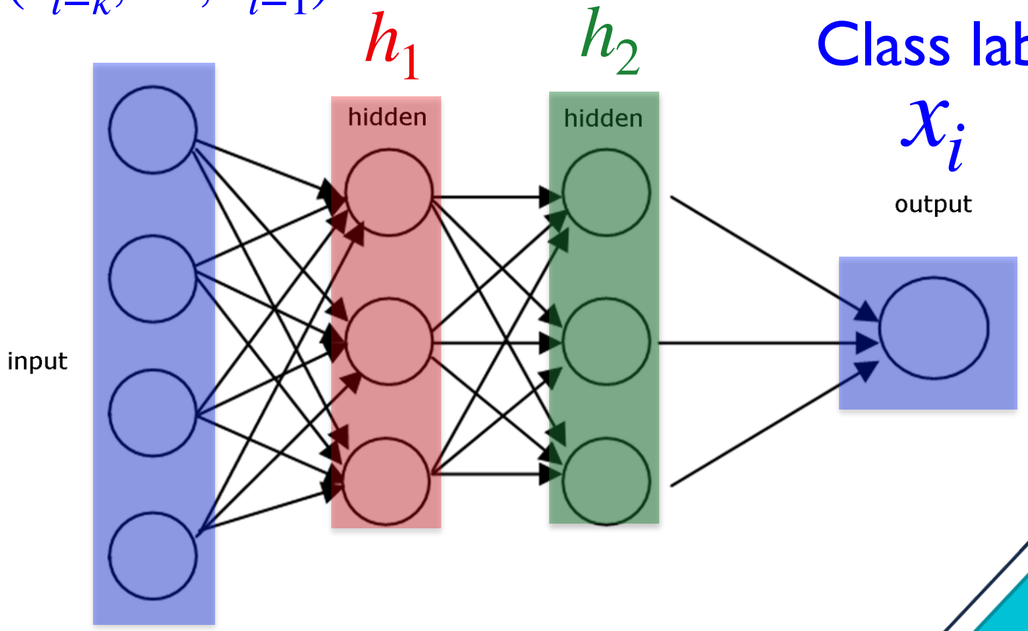

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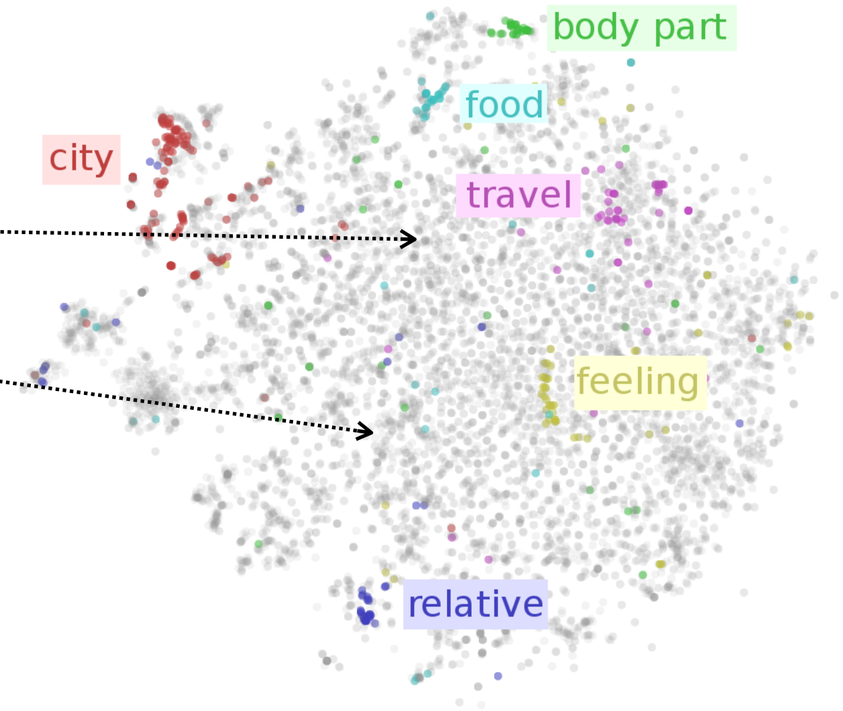

$$p(x_i \mid x_{i-k}, \ldots, x_{i-1})$$

Feature vector

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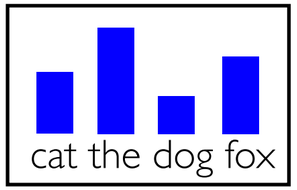

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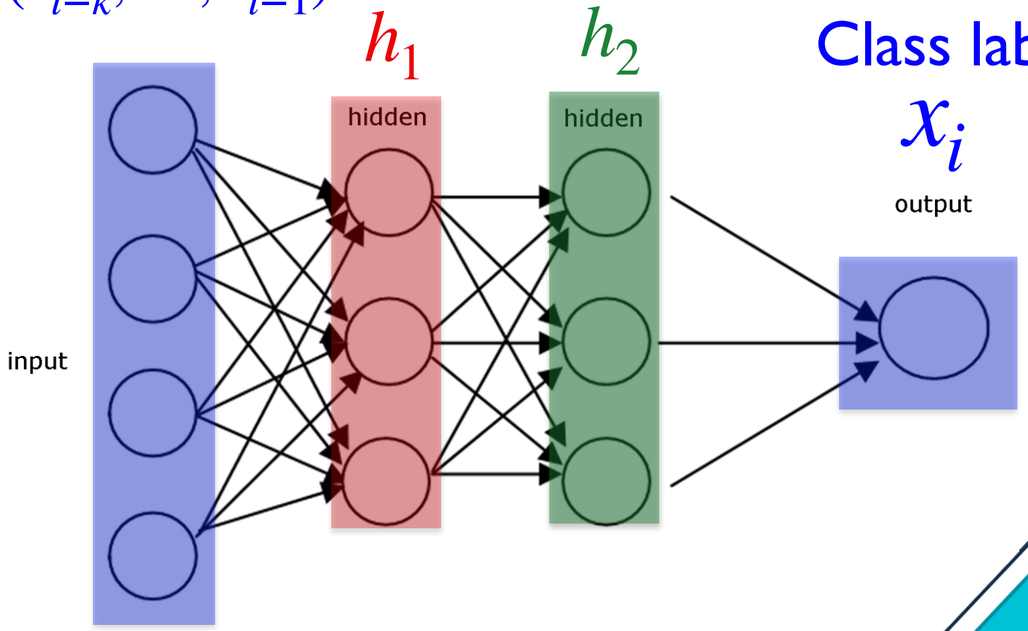
x_i